PROBLEM SET 7 DUE 5.17

1. Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of A-modules. Show that if M_1 and M_3 are finitely presented, then M_2 is finitely presented.

2. Show that the natural map

$$\Big(\prod_{n\geqslant 2} \mathbb{Z}/n\mathbb{Z}\Big) \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \prod_{n\geqslant 2} \big(\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}\big)$$

is not injective.

Hint: $\prod_{n\geqslant 2} \mathbb{Z}/n\mathbb{Z}$ *contains a submodule isomorphic to* $\mathbb{Z}$ *and* $\mathbb{Q}$ *is a flat* $\mathbb{Z}$*-algebra.*

3. Let $\mathfrak{a}$ be a non-finitely generated ideal of a non-Noetherian ring A. Show that there exists a set I such that the natural (surjective) homomorphism

$$\left(\prod_{i\in I} A\right) \otimes_A \big(A/\mathfrak{a}\big) \longrightarrow \prod_{i\in I} A/\mathfrak{a}$$

$\Big($alternatively written as $A^I \otimes_A (A/\mathfrak{a}) \to (A/\mathfrak{a})^I$ $\Big)$ is not injective.

Hint: take I *to be the underlying set of* $\mathfrak{a}$*.*

4. Let A be a ring, M, N be A-modules, and B be a flat A-algebra. Show that if M is of finite presentation, then the natural homomorphism of B-modules

$$\begin{aligned} \mathrm{Hom}_A(M,N) \otimes_A B &\longrightarrow \mathrm{Hom}_B(M \otimes_A B, N \otimes_A B); \\ f \otimes b &\longmapsto bT(f, \mathrm{id}_B) \end{aligned}$$

is an isomorphism.

5. Show that if A is a Noetherian ring, then $A[[x]]$ is a Noetherian ring.

Hint: Imitate the proof of Hilbert's Basis Theorem. For an ideal $\mathfrak{b}$ *of* $A[[x]]$ *and* $n \geqslant 0$*, define* $\mathfrak{a}_n \coloneqq \big\{a \in A \mid \exists$ *a formal power series* $ax^n +$ *higher order terms in* $\mathfrak{b}\big\}$*.*

6. Show that in a Noetherian ring A, every ideal $\mathfrak{a}$ has finitely many minimal prime ideals over $\mathfrak{a}$. In particular, a Noetherian ring has finitely many minimal prime ideals. [1]

Hint: Prove by contradiction. Let Σ *be the set of (proper) ideals which does not have finitely many minimal prime ideals over it. If* $\Sigma \neq \varnothing$*, then under the Noetherian hypothesis,* Σ *has a maximal element* $\mathfrak{a}$*. Obviously,* $\mathfrak{a}$ *can not be prime, so there exist* $x, y \notin \mathfrak{a}$ *such that* $xy \in \mathfrak{a}$*. Show that* $\mathcal{V}(\mathfrak{a}) = \mathcal{V}(\mathfrak{a} + (x)) \cup \mathcal{V}(\mathfrak{a} + (y))$ *where* $\mathcal{V}(\mathfrak{b})$ *denotes the set of prime ideals over* $\mathfrak{b}$*.*

7. Let k be a field and let $A = k[x_1, x_2, \ldots, x_n, \ldots]$ be the polynomial ring of countably infinite variables. Let $\mathfrak{a}$ be the ideal of A generated by all product $x_i x_j$ for $i \neq j$. Show that there are infinitely many minimal prime ideals over $\mathfrak{a}$.

8. Show that in a Noetherian ring A, every ideal $\mathfrak{a}$ contains a power of its radical $\sqrt{\mathfrak{a}}$. In particular, the nilradical of a Noetherian ring is nilpotent. [2]

Find an example of a non-Noetherian ring whose nilradical is not nilpotent.

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[1] See Problem Set **1** Problem **16** for the definition of minimal prime ideals.

[2] See Problem Set **1** Problem **7**.

9. Show that over a Noetherian ring, the direct sum of an arbitrary family of injective modules is injective and the direct product of an arbitrary family of flat modules is flat.

10. Let A be a Noetherian ring and $T \subseteq A$. Show that there exists a finite subset $T_0 \subseteq T$ such that $(T_0) = (T)$. Here, (T_0) and (T) denote the ideals of A generated by T_0 and T respectively.

11. Let M be an A-module and N, K, L be submodules of M such that $K \subseteq L$. Show that there is a short exact sequence of A-modules

$$0 \longrightarrow \frac{L \cap N}{K \cap N} \longrightarrow \frac{L}{K} \longrightarrow \frac{\big(L+N\big)/N}{\big(K+N\big)/N} \longrightarrow 0.$$

12. This exercise is about equivalent definitions of A-algebras and A-algebra homomorphisms. Prove the following statements:

(1) If B is an A-algebra, then the multiplication map $m : B \times B \to B$ is A-bilinear. Conversely, if B is a ring which is also an A-module such that the multiplication map $m : B \times B \to B$ is A-bilinear, then the map $f : A \to B$; $a \mapsto a1_B$ is a ring homomorphism and the A-module structure of B is induced by f.

(2) Let B_1, B_2 be A-algebras; then a set map $\varphi : B_1 \to B_2$ is an A-algebra homomorphism if and only if it is both a ring homomorphism and an A-module homomorphism.

13. Let A be an integral domain and K be its field of fractions. Prove the following statements:

(1) K is a finite A-algebra if and only if $K = A$.
(2) K is a finite type A-algebra if and only if there exists $a \in A$, $a \neq 0$ such that $K = A[\frac{1}{a}]$.

14. Let k be a field. Show that

$$k[x, xy, xy^2, \ldots, xy^n, \ldots]$$

is *not* a finitely generated k-subalgebra of $k[x, y]$.

15. Show that the maximal ideals of $\mathbb{Z}[x]$ have the form $(p, f(x))$ where p is a prime number and $f(x)$ is a monic polynomial in $\mathbb{Z}[x]$ that is irreducible mod p.

Hint: Let $\mathfrak{m}$ be a maximal ideal of $\mathbb{Z}[x]$. The key is to show $\mathfrak{m} \cap \mathbb{Z} \neq 0$. If not, we would have a ring extension $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{Z}[x]/\mathfrak{m}$. Apply Zariski's lemma to $\mathbb{Q} \subseteq \mathbb{Z}[x]/\mathfrak{m}$ and apply Artin-Tate lemma to $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{Z}[x]/\mathfrak{m}$.